S. Smitha Jenifer 2026

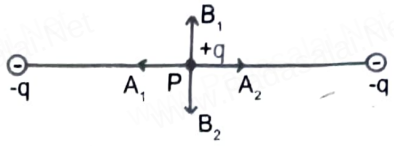
Time : 3.00 hrs. Max. Marks : 70

PHYSICS

PART - I

Answer all questions. Multiple choice questions. **15 x 1 = 15**

1. Two identical point charges of magnitude - q are fixed as shown in the figure below. A third charge +q is placed midway between the two charges at the point P. Suppose this charge +q is displaced a small distance from the point P in the directions indicated by the arrows, in which directions(s) will +q be stable with respect to the displacement?



a) A_1 and A_2 b) B_1 and B_2 c) both directions d) No stable

2. Two points A and B are maintained at a potential of 7V and –4V respectively. The work done in moving 50 electrons from A to B is a) 8.80×10^{-17} J b) -8.80×10^{-17} J c) 4.40×10^{-17} J d) 5.80×10^{-17} J
3. An electric dipole placed in a non-uniform electric field experience
a) both a torque and a net force b) only a force but no torque c) only a torque but no net force
d) no torque and no net force
4. In a large building, there are 15 bulbs of 40W, 5 bulbs of 100W, 5 fans of 80W and 1 heater of 1KW are connected. The voltage of electric main is 220V. The maximum capacity of the main fuse of the building. a) 14A b) 8A c) 10A d) 12A
5. In a wheat stones bridge all the four arms have equal resistance R. If the resistance of the galvanometer is also R, the equivalent resistance of the combination as seen by the battery is
a) R/4 b) R/2 c) R d) 2R
6. A thin insulated wire forms a plane spiral of N =100 light turns carrying a current I = 8mA (milli ampere). The radii of inside and outside turns are a = 50 mm and b = 100 mm respectively. The magnetic induction at the centre of the spiral is a) 5 μT b) 7 μT c) 8 μT d) 10 μT
7. The potential energy of magnetic dipole whose dipole moment is $\vec{P}m = (-0.5\hat{i} + 0.4\hat{j})$ Am² kept in uniform magnetic field $\vec{B} = 0.2\hat{i}$ T a) –0.1 J b) –0.8 J c) 0.1 J d) 0.8 J
8. When the current changes from +2A to –2A in 0.05s, an emf of 8 V is induced in a coil. The co-efficient of self-induction of the coil is a) 0.2 H b) 0.4 H c) 0.8 H d) 0.1 H
9. In an oscillating LC circuit, the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is
a) Q/2 b) $Q/\sqrt{3}$ c) $Q/\sqrt{2}$ d) Q
10. A coil of inductive reactance 31Ω has a resistance of 8Ω. It is placed in series with a condenser of capacitive reactance 25Ω, the combination is connected an a.c. source of 110V, the power factor of the circuit is a) 0.33 b) 0.56 c) 0.64 d) 0.80
11. Which of the following electromagnetic radiations is used for providing information about the structure of atomic nuclei? a) microwave b) gamma rays c) X-rays d) infrared
12. Fraunhofer lines are an example of.....spectrum.
a) line emission b) line absorption c) band emission d) band absorption
13. In an electromagnetic wave travelling in free space the rms value of the electric field is 3 Vm^{-1}. The peak value of the magnetic field is a) 1.414×10^{-8} T b) 1.0×10^{-8} T c) 2.828×10^{-8} T d) 2.0×10^{-8} T
14. The speed of light in an isotropic medium depends on,
a) its intensity b) its wavelength c) the nature of propagation d) the motion of the source w.r.t medium
15. The radius of curvature of curved surface at a thin planoconvex lens is 10 cm and the refractive index is 1.5. If the plane surface is silvered, then the focal length will be,
a) 5 cm b) 10 cm c) 15 cm d) 20 cm

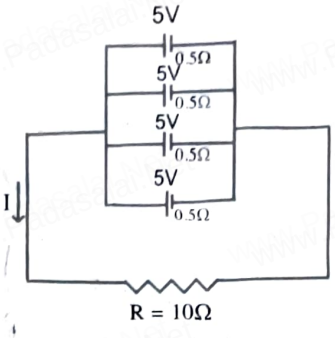
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PART - II

Answer any six of the following. Question Number 24 compulsory. **6 x 2 = 12**

16. Define 'electric flux'.
17. Define electrical resistivity.
18. For the given circuit, find Equivalent emf and Equivalent internal resistance.



19. What is meant by hysteresis?
20. What is magnetic susceptibility?
21. State Lenz's law.
22. Give two uses of IR radiation.
23. Why do clouds appear white?
24. Pure water has refractive index '1.33. What is the speed of light through it?

PART - III

Answer any six of the following. **6 x 3 = 18**
Question Number 33 compulsory.

25. Derive an expression for electrostatic potential due to a point charge.
26. Dielectric strength of air is 3×10^6 Vm^{-1}. Suppose the radius of a hollow sphere in the Van de Graff generator R = 0.5 m, calculate the maximum potential difference created by this Van de Graaff generator.
27. State Kirchhoff's rules.
28. Explain how is a galvanometer converted into an ammeter.
29. What are step-up and step-down transformers?
30. How is Eddy current produced? How do they flow in a conductor?
31. Write down the properties of electromagnetic waves.
32. What is optical path? Obtain the equation for optical path.
33. The repulsive force between two magnetic poles in air is 9×10^{-3} N. If the two poles are equal in strength and are separated by a distance of 10cm, calculate the pole strength of each pole.

PART - IV

Answer all the questions. **5 x 5 = 25**

34. a) Calculate the electric field due to a dipole on its equatorial plane. **(OR)**
    b) Derive the mirror equations.
35. a) What is emission spectrum. Explain the types of emission spectrum. **(OR)**
    b) Obtain an expression for motional emf from Lorentz force.
36. a) Explain the determination of the internal resistance of a cell using voltmeter. **(OR)**
    b) Derive the expression for the force on a current-carrying conductor in a magnetic field.
37. a) Show mathematically that the rotations of a coil in a magnetic field over one rotation induces an alternating emf of one cycle. **(OR)**
    b) How the emf of two cells are compared using potentiometer?
38. a) Derive the expression for resultant capacitance, when capacitors are connected in series and in parallel. **(OR)**
    b) (i) Write the difference between the electric field (from Coulombs law) and magnetic field (from Biot-savart law)
    (ii) Compute the magnitude of the magnetic field of a long, straight wire carrying a current of 1A at distance of 1m from it.

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